

Falcon 2.0T camshaft timing over-advanced bank 1 from sticking VVT solenoid

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Area: bulletins

1 Condition

Falcon 2.0T vehicles may store DTC P0011 (Camshaft Position A - Timing Over-Advanced or System Performance Bank 1) with the malfunction indicator lamp illuminated. Drivers may notice a rough idle, hesitation on light acceleration, and occasionally a no-restart-until-cooldown stall after a hot soak. The condition is intermittent at first and becomes more consistent as the affected component degrades. For the location and operation of the timing actuator, refer to Falcon Engine Service Manual :: Variable Valve Timing System .

2 Cause

DTC P0011 on these engines is caused by variable valve timing solenoid VVT-3360 sticking in a partially open position, which over-advances camshaft phasing on bank 1 beyond the commanded target. Oil varnish accumulation on the spool of VVT-3360 prevents it from returning fully, so the ECM detects a phase error and sets the code. Background on phasing control is covered in Harrier Engine Service Manual :: Timing Sub-System , and a comparable solenoid failure mode is documented in Lumen Engine Service Manual :: Valve Timing Sub-System .

3 Repair Procedure

Replace variable valve timing solenoid VVT-3360 following procedure PROC-VVT-SOL, inspecting the solenoid screen for debris and confirming clean oil flow before installation. Step-by-step torque and indexing instructions are provided in Falcon Engine Service Manual :: VVT Solenoid Replacement ; equivalent procedures for reference are Harrier Engine Service Manual :: Variable Valve Timing Solenoid Replacement and Lumen Engine Service Manual :: Variable Valve Timing Solenoid Service . After replacing VVT-3360, clear P0011, perform a relearn, and road test to confirm stable phasing under load.